

JAESEOK HWANG

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Ph.D. in Applied Economics | Treasury-Market Monitoring · Scenario Analytics · Decision-Risk Frameworks

RESEARCH & ANALYTICS EXPERIENCE

University of Illinois Urbana-Champaign

Sep 2020 – Present

Research Assistant, DIFM Project (USDA-funded; 16-university consortium)

- Built end-to-end API and SQL data pipelines for a USDA-funded decision-support initiative with approximately \$70M in total project scale, ensuring integrity of multi-source inputs used for applied analytics and reporting.
- Translated complex behavioral and economic scenarios into quantitative decision frameworks using Python, R, and Excel to evaluate expected outcomes, downside risk, and sensitivity to changing external conditions over 1M+ observations.
- Owned end-to-end analytical work streams spanning data preparation, modeling, visualization, and written synthesis; collaborated with engineers, economists, and project leads to deliver decision-relevant outputs on tight timelines.

ANALYTICS PROJECTS [Project page](#)

Independent Treasury & Macro Monitoring Project (Ongoing)

- Built an ongoing real-time monitoring workflow for Treasury yields, inflation compensation, Fed expectations, credit spreads, FX, oil, and key U.S. macro releases to track changing market regimes and policy-sensitive repricing.
- Current work emphasizes Treasury-market and macro analysis following the recent U.S.-Iran conflict, focusing on how geopolitical shocks may alter growth, inflation, funding conditions, and broader economic regime dynamics, and summarizing the results through concise charts and scenario notes.

Information Transferability and Economic Risk Without Structural Validation (2024; submitted)

- Showed that predictive fit alone is not sufficient when structural conditions shift, highlighting the importance of testing assumptions before scaling model-based recommendations; used leave-one-location-out validation with XGBoost and Random Forest under heterogeneous operating environments.

Decomposing Profitability Risk: Decision Error vs. Exogenous Shocks (2025)

- Developed a framework separating controllable decision error from external shocks to clarify whether deteriorating outcomes reflect model misspecification or macro volatility, using hierarchical models with bootstrap-based uncertainty simulation.

QUANTITATIVE METHODS & TOOLS

- **Analytics:** Time-series analysis, scenario analysis, stress testing, Bayesian inference, causal inference, Monte Carlo and bootstrap simulation.
- **Treasury & Macro Context:** U.S. data-release monitoring, Fed interpretation, Treasury-curve moves, real yields vs. breakevens, credit-spread monitoring, and macro regime assessment.
- **Tools:** Python, R, SQL, Excel, PowerPoint, Git, FRED/BLS/BEA/EIA public-data workflows, API-to-reporting pipelines, and data visualization dashboards.

EDUCATION

University of Illinois at Urbana-Champaign

Expected Aug 2026

Ph.D. in Applied Economics

Focused coursework: Financial Analysis, Macroeconomics, Advanced Spatial Econometrics (received all A's)

University of Illinois Urbana-Champaign

Aug 2020

M.A. in Applied Economics

Sogang University (Korea)

Aug 2013

B.S. Economics